Anna Haensch

Contact

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Academic & Professional Appointments

University of Wisconsin, Madison, WI
Research Associate Professor, Data Science Institute, Sept. 2025 – present.
Associate Director, Digital Scholarship Hub, Sept. 2025 – present.

United States Senate, Office of U.S. Senator Elizabeth Warren, Washington, DC
Legislative Fellow for Artificial Intelligence, Sept. 2024 – August 2025.

Tufts University, Medford, MA
Senior Data Scientist, Tufts Institute for AI (formerly Data Intensive Studies Center), Jan. 2021 – Aug. 2025.
Secondary Appointment, Department of Mathematics, July 2021 – Aug. 2025.
Provost Lecturer, Fletcher Graduate School of Global Affairs, July 2021 – Aug. 2025.

Tagup Inc., Somerville, MA
Research Data Scientist, Sept. 2019 – Dec. 2020.

Duquesne University, Pittsburgh, PA
Associate Professor, Department of Mathematics & Computer Science, Aug. 2020 – May 2021.
Assistant Professor, Department of Mathematics & Computer Science, Aug. 2013 – Aug. 2020.

Max Planck Institute for Mathematics, Bonn, Germany
Visiting Researcher, Jan. 2018 – May 2018; Jan. 2014 – May 2014.

National Public Radio, Washington, D.C.
AAAS-AMS Mass Media Fellow, May 2013 – Aug. 2013.

Education

Wesleyan University, Middletown, CT
Ph.D. Mathematics, May 2013.

State University of New York at New Paltz, New Paltz, NY
B.S. Mathematics, May 2007.

Recent Grant Funding

- (Under Review) Co-I, NASA, *Informing Health and Environmental Impacts of Data Centers with Satellite Data*, $150,000, (January 2027 - December 2027)
- (Under Review), Senior Personnel, NSF, *TechAccess: AI-Ready America: Wisconsin Coordination Hub: Advancing Statewide Artificial Intelligence Literacy, Workforce Development, and Adoption Across Sectors* $3,000,000, (January 2027 - December 2029)

- (Under Review), Advisory Board Member, Sloan Foundation, *Adaptive Community Benefit Agreements for Data Center Load Growth, Electricity Affordability, and Local Energy Governance*, $250,000, with PI. E. Tronci, (January 2027 - December 2029).
- (Awarded - Current) PI, Nelson Institute for Environmental Studies, *Co-Development of a Framework for Hyperscale Data Center Community Benefits Agreements*, $17,000, with Co-PI Esther Kang, (July 2026 - June 2027)
- (Awarded - Current) Co-PI, National Science Foundation Partnership for International Research and Education (NSF-PIRE), *Multi-Domain, Multi-Scale, Policy-Aware Digital Twin for Offshore Wind Energy Infrastructure*, $1,498,144, with PI B. Moaveni, Co-PI U. Khan, and Co-PI H. Ebrahimian, (January 2023 – December 2026 estimated) Award OISE-2230630.
- (Awarded) Key Personnel, Department of Defense Small Business Innovation Research (DoD-SBIR), *LAV25 Logistics Optimization using Machine Learning*, $1,593,070, with PI William Vega Brown, (May 2020 – November 2021) Award N68335-20-F-0459.
- (Awarded) PI, National Science Foundation Division of Mathematical Sciences (NSF-DMS), *Workshop on Automorphic Forms and Related Topics*, $14,987, with Co-PIs Ben Kane, Djordje Milićević, and Howard Skogman, (March 2019 - February 2020) Award DMS-1854113.

PREPRINTS *(* denotes papers where authors are ordered alphabetically)*

1. **A. Haensch**, E.M. Tronci, Hyperscale Data Centers and Wind Energy: Cross-Sector Lessons in Siting, Permitting, and Community Opposition, under review, preprint at http://dx.doi.org/10.2139/ssrn.6506945.

PUBLICATIONS *(* denotes papers where authors are ordered alphabetically)*

34. M. Alikhani, J. Kim, **A. Haensch**, S. Imai, S. Booth, A.L. Stith, A. Bankston Bridging AI Research and Policy: A Co-Production Framework and the Case for Policy Transfer Offices, *Proceedings of the Ninth AAAI/ACM Conference on AI, Ethics, and Society*, 9(1), 2026. DOI: https://doi.org/10.1609/aies.v9i1.43128
33. **A. Haensch**, E. M. Tronci, B. Moaveni, Uncertainty Quantification of Bending Moments with Characterization of Strain Measurement Error on Offshore Wind Turbines, *J. Phys.: Conf. Ser.* **3224** 062039, 2026. DOI: 10.1088/1742-6596/3224/6/062039
32. **A. Haensch**. The Story Your Ballot Doesn't Tell: The Mathematics of Single-Winner Elections, *American Mathematical Society Feature Column*, May 2026.
31. **A. Haensch**, E. M. Tronci, A. Banerjee, V. Valencia, B. Moaveni. A Safety-Aware Framework for Offshore Wind Turbine Maintenance, *Wind Energy*, 28(12), 2025, p. e70065. DOI: https://doi.org/10.1002/we.70065
30. **A. Haensch**, C. Kelling, BlockR: An Areal Spatial Anonymization and Visualization Tool, *Transactions in GIS*, 29(4), 2025, 1 - 11. DOI: https://doi.org/10.1111/tgis.70070
29. **A. Haensch**, Becoming a Data Scientist: An Impractical Guide, *The American Mathematical Monthly*, 132(1), 2024, 6 – 13. DOI: 10.1080/00029890.2024.240906428.
28. * J. W. Bourgeois, **A. Haensch**, S. Kher, D. Knox, G. Lanzalotto, and T.A. Wong, How to Use Causal Inference to Study Use of Force, *CHANCE*, 37:4, 2024, 6-10. DOI:10.1080/09332480.2024.2434435
27. C. Kelling, **A. Haensch**, A. Mendible, S. Brooks, A. Wiedemann, M. Aminian, W. Hasty, J. Higdon. Data Collection and Analysis for Small-Town Policing: Challenges and Recommendations, *Statistics and Public Policy*, 1-21, July 2024. DOI:10.1080/2330443X.2024.2379270.

26. **A. Haensch**, D. Gordon, K. Knudson, J. Cheng. A Multi-Method Data Science Pipeline for Analyzing Police Service, *The American Statistician*, 1-18, July 2024. DOI:10.1080/00031305.2024.2374275

25. * C. Börgers, N. Dragovic, **A. Haensch**, A. Kirshtein. Particle Method for Continuous Hegselmann-Krause Opinion Dynamics. In *Complex Networks & Their Applications XII* (pp. 1-13). Springer Nature Switzerland AG. 2024. DOI: 10.1007/978-3-031-53499-7_37 final version at arXiv:2211.06265.

24. **A. Haensch**, D. Deitsch. An Equity-Aware Recommender System for Curating Art Exhibits Based on Locally-Constrained Graph Matching, ArXiv arXiv:2207.14367.

23. **A. Haensch**, E. Tronci, B. Moynihan, B. Moaveni. Regularized Hidden Markov Modeling with Applications to Wind Speed Predictions in Offshore Wind, *Mechanical Systems and Signal Processing*, Vol. 211, 2024, 111229.

22. * C. Börgers, N. Dragovic, **A. Haensch**, Political Centrism and Extremism: A Mathematical Analysis, *SIAM News*, January 19, 2024.

21. * C. Börgers, N. Dragovic, **A. Haensch**, A. Kirshtein, L. Orr. ODEs and Mandatory Voting, CODEE Journal special issue on *Engaging the World: Differential Equations Influence Public Policy*, Vol. 17, Article 11, 2024.

20. E. Tronci, **A. Haensch**, G. Georgalis, B. Moaveni. Uncertainty Quantification of Bending Moments with Characterization of Strain Measurement Error on Offshore Wind Turbines, extended abstract accepted to IMAC-XLII *Uncertainty Quantification in Dynamics*, Jan. 2024.

19. **A. Haensch**, E. Tronci, B. Moaveni, E. Hines. Multi-Factor Decision Framework for Offshore Wind Turbine Maintenance, extended abstract accepted to IMAC-XLII *Structural Modelling and Condition Assessment*, Jan. 2024.

18. * C. Börgers, B. Boghosian, N. Dragovic, **A. Haensch**. A blue sky bifurcation in the dynamics of political candidates, *The American Mathematical Monthly*, 131:3, 2023, 225 – 238 DOI:10.1080/00029890.2023.2286221 final version at arXiv:2302.07993.

17. **A. Haensch**, N. Dragovic, C. Börgers, B. Boghosian. A geospatial bounded confidence model including mega-influencers with an application to Covid-19 vaccine hesitancy, *The Journal of Artificial Societies and Social Simulations*, **26** (1) 2023, 8.

16. **A. Haensch**. Review: Data Feminism, *The American Mathematical Monthly*, **129:5** (2022), 496 – 500. DOI: 10.1080/00029890.2022.2044214

15. **A. Haensch**, K. Knudson. Python for Global Applications: Teaching scientific Python in context to law and diplomacy students, *Proceedings of the 21st Python in Science Conference*, 2022, 59 – 64, final version at https://conference.scipy.org/proceedings/scipy2022/anna_haensch.html.

14. **A. Haensch**, I. Ljungberg, U. Khan, B. Moaveni. Monitoring of Offshore Wind Turbines Using Measured Accelerations and Hidden Markov Models with Physics-Based Initialization, extended abstract in conference proceedings IMAC-XL *Data Science for Advanced Manufacturing*, Society for Experimental Mechanics, 2022.

13. **A. Haensch**. Reflections on Hyperbolic Space, *Snapshots of modern mathematics from Oberwolfach* **7** (2021), 1 – 10. DOI: 10.14760/SNAP-2021-007-EN .

12. * M. Dutour Sikirić, **A. Haensch**, J. Voight and W. van Woerden. A canonical form for positive definite matrices, Proceedings of the *Fourteenth Algorithmic Number Theory Symposium, ANTS-XIV*, Mathematical Sciences Publishers, 2020, final version at arXiv:2004.14022.

11. * A. G. Earnest, **A. Haensch**. Classification of one-class spinor genera for quaternary quadratic forms, *Act Arith.* **91** 3 (2019) 259 – 287, final version at arXiv:1803.03028.

10. * **A. Haensch**, B. Kane. An algebraic and analytic approach to spinor exceptional behavior in translated lattices, *Automorphic Forms and Related Topics, Contemp. Math.*, **732**, 2019, Amer. Math. Soc., Providence, RI final version pdf available here.

9. * A. G. Earnest, **A. Haensch**. Completeness of the list of spinor regular ternary quadratic forms, *Mathematika*, **65** (2019), 213–235, final version at arXiv:1711.05811.

8. **A. Haensch**. Review: Foolproof and Other Mathematical Meditations, *Math Horizons*, **25:4** (2018), 29–29. DOI: 10.1080/10724117.2018.1434292

7. * **A. Haensch**, B. Kane. Almost universal ternary sums of polygonal numbers, *Res. number theory* (2018) 4: 4, final version at https://doi.org/10.1007/s40993-018-0098-x.

6. **A. Haensch**. The Blog on Math Blogs, *Notices of the American Mathematical Society*, **63:6** (2016), 643–644. DOI: http://dx.doi.org/10.1090/noti1387

5. * A. Feaver, **A. Haensch**, J. Liu, G. Nebe. On Kneser-Hecke operators for codes over finite chain rings, *Directions in Number Theory: Proceedings of the 2014 WIN3 Workshop*, Association for Women in Mathematics Series, Springer-Verlag, (2016).

4. **A. Haensch**. A characterization of almost universal ternary inhomogeneous quadratic polynomials with conductor 2, *J. Number Theory*, **156** (2015), 247–262.

3. **A. Haensch** A characterization of almost universal ternary quadratic polynomials with odd prime power conductor, *J. Number Theory*, **141** (2014), 202–213.

2. **A. Haensch**. My Summer at NPR, *Notices of the American Mathematical Society*, **60:11** (2013), 1477–1478. https://www.ams.org/notices/201311/rnoti-p1477.pdf

1. * W. K. Chan, **A. Haensch**. Almost universal ternary sums of squares and triangular numbers, *Quadratic and Higher Degree Forms*, Developments in Mathematics, Springer-Verlag, (2013).

0. * K. Doerksen, **A. Haensch**. Primitive prime divisors in zero orbits of polynomials, *INTEGERS: The Online Journal of Combinatorial Number Theory*, **12** (2012).

Selected Invited Talks

- Invited Speaker, *The Mathematics Of Why You Can't Win: Opinion Dynamics, Electoral Strategy, and Wealth Concentration*, Banff International Research Station, March 2026.
- Invited Speaker, *Data Centers in 2026*, Understanding AI: A Forum for Local & Tribal Governments, UW-Extension, Wisconsin Dells, Wisconsin, March 2026.
- Invited Speaker, *Is AI Coming For Your Job? And Other Mathematical Questions*, Denison University, Granville, OH, March 2026.
- Invited Speaker, *Data, Democracy, & The Role of the Mathematician*, Joint Mathematics Meetings, Washington, D.C., January 2026.
- Invited Speaker, *Police Accountability in the Age of Algorithms*, MIT Seminar on the Digital Humanities, Cambridge, MA, October 2025.
- Plenary Speaker, *Becoming a Data Scientist: An Impractical Guide*, Math For All Conference, Tulane University, New Orleans, LA, April 2024.
- Invited Speaker, *Data Science for Police Accountability*, X-SIG Seminar Series, Gettysburg College, Gettysburg, PA, Feb. 2023.

- Featured Speaker, *From Riemann Zeta to Big Data: A journey through mathematics and the lessons learned along the way*, Graduate Research Opportunities for Women (GROW) 2022, Duke University, Durham, NC, Oct. 2022.
- Selected Speaker, *Python for Global Applications: Teaching scientific Python in context to law and diplomacy students*, SciPy: Scientific Computing with Python Conference, Austin, TX, July 2022.
- Featured Speaker, *An Equity-Aware Recommender System for Curating Art Exhibits Based on Locally-Constrained Graph Matching*, Data4Justice Conference, Institute for the Quantitative Study of Inclusion, Diversity & Equity, *virtual*, April 2022.
- Plenary Speaker, *Some results on class numbers for quadratic forms*, Upstate Number Theory Conference, University of Buffalo, Buffalo, NY, April 2018.
- Invited Speaker, *Quadratic forms and the representation problem*, Oberseminar, Max Planck Institute for Mathematics, Bonn, Germany, March, 2018.

Selected Professional Service

Editorial & Advisory

- Member at Large, American Mathematical Society Committee on Science Policy, January 2027 - present.
- Advisory Board Member, Center for Humanistic Inquiry into AI and Uncertainty, May 2026 - present.
- Associate Editor, The American Mathematical Monthly, January 2026 - present.
- Expert Advisor, Library of Congress Federal Research Division and John W. Kluge Center, January 2026.
- Member at Large, Joint AMS-ASA-MAA-SIAM Data Committee, January 2025 - present.
- Guest Editor, The American Mathematical Monthly, Special Issue on Data Science, 2025.
- Proposal Review Panelist, National Science Foundation, 2022, 2023.
- External Reviewer, Quantitative Analysis Center, Wesleyan University, 2022.

Conferences & Panels

- Program Committee, *8th Natural Legal Language Processing Workshop*, Budapest, Hungary, October 2026.
- Senior Program Committee Member, *AAAI/ACM Conference on AI, Ethics, and Society*, Malmö, Sweden, October 2026.
- Organizer, *AI & the Workforce: Planning a Career in the Age of AI, Presentations, Panel Discussion, and Workshop*, UW-Madison, Madison, Wisconsin, March 2026.
- Panelist, *Bringing Math to Congress: Advocating for Change*, Joint Mathematics Meetings, Washington, D.C., January 2026.
- Program Chair, *SciPy: Scientific Computing with Python Conference*, Tacoma, WA, 2024.
- Facilitator and Panelist, *AI, Design and Afro-Nowism, Tufts University*, Medford, MA, 2024 (with D.Martinez).
- Track Chair for Digital Humanities, *SciPy: Scientific Computing with Python Conference*, Austin, TX,2023.
- Organizer, *AMS Spring Sectional Meeting - Special Session on Mathematics of Data Science*, Tufts University, Medford, MA, 2022 (with A. Tasissa, J. Murphy, A. Patra, and V. Maroulas).
- Organizer, *Art Datathon*, Tufts University, Medford, MA 2024 (with D. Deitsch).

- Organizer and Moderator, *A Conversation with the New York Times*, Tufts University, Medford, MA 2021.
- Organizer, *33rd Automorphic Forms Workshop*, with funding from the National Science Foundation Award DMS-1854113, Duquesne University, Pittsburgh, PA, 2019 (with B. Kane, D. Milićević, and H. Skogman).
- Organizer, *Sage Days 90: Women in Sage*, with funding from Microsoft Research and the Beatrice YorkmarkFoundation, Harvey Mudd College, Claremont, CA, 2017 (with A. Deines, B. Thompson, U. Whitcher).

Pedagogical Interests

My approach to teaching data science is grounded in practical and applied data science with an emphasis on the social and ethical considerations of new technology. Seating the technical aspects of data science in surrounding literature from the social sciences and current events is a critical cornerstone of my courses. I've taught various level courses to students from the humanities and the sciences using both R and Python. I've advised undergraduate and graduate researchers in projects involving data mining, web scraping, optical character recognition, visualization, analysis, and algorithm programming and optimization.

Teaching Experience

Selected Courses Taught *(100 level are advanced undergraduate courses, 200 level are graduate courses)*

- *Data Science and AI Policy* - A foundational course focused on the full data life-cycle including the generation, collection, processing, analysis, and management of data. The course also presents recent advances in artificial intelligence and the opportunities and threats that these pose as well as related legislative actions. *GMA-D258 at Tufts University*
- *Data Science Practice for Law and Diplomacy* - A foundational course in data science covering data exploration, data visualization, and data modeling. Coding is carried out in Python and the technical work is backed by readings on the social, political, and global imacts of big data. *DHP-D258 at Tufts University*
- *Mathematical Models in Social Context* - A survey course focused on Science, Technology & Society literature on models and algorithms. The course addresses concepts like model selection and impact, ethical breakdowns in big data and algorithms, and knowledge in the context of big data. *STS-10 at Tufts University*
- *Foundations of Data Analysis in R* - A beginner technical course establishing a unifying framework for how data should be used to answer and explore different questions. The course provides an introduction into the ways that data are used and analyzed across the different disciplines within the arts, humanities, and sciences using popular R libraries. *DATA-200 at Tufts University*
- *Mathematical Aspects of Data Analysis in Python* - A technical course in mathematical data science with an emphasis on theory, covering topics like PCA, clustering and density methods, neural networks and spectra graph theory. The course will also highlight important applications and students will have the opportunity to program some standard algorithms and use explore existing Python libraries. *MATH-123 at Tufts University.*
- Occasional short workshops: *Data Visualization with Python*, *How to Set up a Local Computing Environment*, *How to work with an API*, and others.

Students Advised & Mentored

- Aidan Banerjee, Undergraduate Research Assistant, Jan. 2024 – present.
- Bridget Moynihan, Graduate Research Assistant, May 2022 – present.
- Mackenzie McPike, Graduate Research Assistant, Sept. 2023 – Dec. 2023.
- Rosie Rong, Graduate Research Assistant, May 2022 – May 2023.

- Isabel Ljungberg, Master's Thesis Advisor, degree completed Dec. 2021.
- Tyler Gaona, Undergraduate Research Assistant, Sept. 2015 – May 2017.
- C. Thomas Dean, Undergraduate Research Assistant, Sept. 2014 – Dec 2014.

Media Outreach

- Guest, *Sustain Dane Podcast*, Madison, WI, April 2026.
- Moderator, *A Conversation with the New York Times*, Tufts University, Medford, MA, Sept. 2021.
- Guest, *As It Happens*, CBC Radio One, June 2017.
- Guest, *Press Play with Madeleine Brand*, KCRW the NPR station out of Santa Monica, CA, March 2017.
- Guest, *Essential Pittsburgh*, WESA Pittsburgh's NPR News Station, May 2016.
- Podcast Co-Host, *The Other Half*, a podcast about math, available on iTunes, May 2015-May 2016.
- Editor/Contributor, AMS Blog on Math Blogs, Feb. 2015– Aug. 2019.